

GALUUT, Mongolia

WMO: 442840

Lat: 46.7020N

Lon: 100.1430E

Elev: 2126

StdP: 78.26

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.5	-34.2	-39.5	0.1	-35.3	-37.3	0.1	-33.2	7.8	-16.3	6.0	-18.3	0.6	0	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	25.2	13.6	23.4	12.6	21.9	11.8	15.7	22.0	14.4	20.6	13.3	19.3	3.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	12.5	18.3	12.1	11.4	17.3	10.8	10.4	15.9	52.4	22.0	48.0	20.7	44.3	19.4	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	7.8	6.6	-38.4	28.3	2.8	1.6	-40.4	29.5	-42.1	30.5	-43.6	31.4	-45.7	32.5
			-38.4	16.9	2.8	2.8	-40.5	18.9	-42.1	20.6	-43.7	22.1	-45.8	24.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.0	-24.7	-20.6	-10.3	0.6	7.0	12.9	15.4	13.3	6.7	-2.4	-13.4	-21.7		
	DBStd	14.77	4.68	5.50	6.60	5.01	4.30	3.43	3.00	2.96	4.19	4.82	6.11	4.72		
	HDD10.0	5153	1076	856	628	283	111	13	1	7	112	383	701	983		
	HDD18.3	7802	1334	1089	886	533	352	166	100	159	349	642	951	1242		
	CDD10.0	407	0	0	0	1	18	99	168	108	13	0	0	0		
	CDD18.3	13	0	0	0	0	0	2	8	2	0	0	0	0		
	CDH23.3	161	0	0	0	0	2	36	92	31	0	0	0	0		
	CDH26.7	12	0	0	0	0	0	2	8	2	0	0	0	0		
	WSAvg	1.7	0.6	0.9	1.8	2.7	2.9	2.3	2.0	1.7	1.7	1.5	1.0	0.8		
Precipitation	PrecAvg	211	2	2	4	8	14	39	64	56	17	7	3	2		
	PrecMax	459	10	7	18	48	112	138	120	141	53	39	16	9		
	PrecMin	83	0	0	0	0	1	3	17	7	0	0	0	0		
	PrecStd	69	2	2	5	9	19	27	28	30	12	8	4	2		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-9.2	-3.2	8.6	17.0	23.1	26.7	27.8	26.4	21.1	13.6	5.1	-6.2		
		MCWB	-11.5	-6.2	0.8	7.6	10.1	14.4	15.3	14.0	9.8	4.7	0.2	-8.6		
	2%	DB	-12.4	-6.9	5.5	14.3	20.4	24.1	25.8	23.7	18.8	11.0	1.5	-10.0		
		MCWB	-13.9	-8.8	-1.3	5.0	9.0	12.8	14.1	12.3	8.9	3.9	-2.3	-11.6		
	5%	DB	-15.2	-9.6	2.9	12.0	18.1	22.4	24.0	21.8	16.8	8.6	-1.4	-12.3		
		MCWB	-16.0	-11.1	-2.5	3.6	7.9	11.6	13.4	11.8	8.0	2.2	-4.3	-13.5		
	10%	DB	-17.6	-11.9	0.1	9.7	16.0	20.6	22.3	20.1	14.8	6.2	-3.9	-14.4		
		MCWB	-18.2	-13.1	-3.8	2.4	6.6	10.9	12.5	11.1	7.0	0.8	-6.3	-15.3		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.3	-5.9	1.3	9.2	12.4	16.7	17.9	16.6	11.5	6.2	0.5	-8.5		
		MCDB	-9.5	-3.8	7.2	15.2	20.5	24.3	23.9	22.7	18.0	11.8	4.6	-6.5		
	2%	WB	-13.9	-8.8	-0.8	6.1	10.4	14.8	16.3	14.5	10.0	4.2	-2.1	-11.4		
		MCDB	-12.6	-7.0	4.4	12.8	17.8	21.7	22.6	19.7	16.5	9.9	1.4	-10.1		
	5%	WB	-16.0	-11.0	-2.2	4.2	9.0	13.2	15.0	13.2	8.8	2.6	-4.3	-13.4		
		MCDB	-15.2	-9.6	2.5	10.9	16.4	19.4	21.4	18.9	15.6	8.0	-1.4	-12.3		
	10%	WB	-18.2	-13.0	-3.8	2.6	7.5	12.1	13.8	12.1	7.7	1.0	-6.2	-15.3		
		MCDB	-17.6	-11.9	0.1	9.2	14.7	18.2	20.2	18.3	13.7	5.8	-3.9	-14.4		
Mean Daily Temperature Range	5% DB	MDBR	11.1	13.3	14.2	12.6	12.7	12.2	11.3	11.9	13.2	13.6	12.4	10.5		
		MCDBR	12.8	14.6	15.3	15.6	16.2	15.1	13.8	14.5	15.5	16.2	14.9	12.5		
		MCWBR	11.5	12.8	10.2	9.1	8.8	7.4	6.1	6.5	8.2	10.2	11.5	11.2		
	5% WB	MCDBR	12.7	14.4	14.6	14.7	14.5	12.9	12.0	11.7	13.6	15.4	14.5	12.3		
		MCWBR	11.5	12.8	10.2	9.3	9.2	7.7	6.6	6.3	7.9	10.3	11.3	11.1		
Clear-Sky Solar Irradiance	taub		0.215	0.241	0.282	0.322	0.335	0.343	0.342	0.319	0.287	0.263	0.234	0.216		
	taud		2.411	2.355	2.285	2.266	2.288	2.358	2.408	2.447	2.487	2.474	2.441	2.406		
	Ebn at Noon		923	952	947	933	928	919	916	927	935	914	885	878		
	Edh at Noon		66	87	110	125	127	119	112	103	90	77	64	59		
All-Sky Solar Radiation	RadAvg		1.99	3.07	4.55	5.90	6.75	6.62	6.28	5.69	4.91	3.39	2.15	1.63		
	RadStd		0.13	0.14	0.18	0.24	0.45	0.39	0.34	0.30	0.23	0.18	0.12	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-2.85	N/A	N/A	-1.86	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	-2.48	N/A	-0.82	-0.90	N/A	N/A	N/A	N/A

Nomenclature: See separate page